GENERATIVE NETWORKS FOR EMULATING SYNTHETIC SKY IMAGES

Emulating astronomical images using Deep Generative models conditioned with Gaussian Processes or Autoregressive flows.

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ABSTRACT

Robust generation of high-fidelity data is an essential component of large astronomical survey analysis. While expensive simulations are typically used for creating synthetic data, recent development of fast emulators using machine learning techniques like Generative models – Variational Autoencoders[1], Generative Adversarial Networks[2] or Gaussian Processes – have made high precision predictions of cosmological functions possible. This projects deals with a natural progression of the above for emulating astronomical images using Deep Generative models. The project involves creating training images using GalSim, a software library for generating images of stars and galaxies, tuning a relatively few number of physical parameters on a space-filling scheme. A second dataset, made of real galaxy images from Cosmic Evolution Survey – COSMOS[3, 4] –, is also considered in this work. Ensemble of generative networks, Principal Component Analysis, Gaussian Processes, Variational Autoencoders and Masked Autoregressive Flow[5], is trained at interpolating values of input parameters. Independent measurement pipelines are applied to validate the emulated images beyond visual confirmation and hold-out tests. The network will be made more physics-aware based on this preliminary results and is extended to emulate realistic synthetic images tuned for other surveys and catalogs. Generative models designed in this work are compared at generating galaxy images and Deep generative models as Variational Autoencoders outperform machine learning techniques such as Principal Component Analysis for analytical and real images. An effort will be made in future work to improve current models performances and interpolate physical parameters to extend emulation to other surveys, as Illustris/TNG simulations.

***K*eywords** Galaxy emulation Neural Networks Generative models Variational Autoencoder

1 Introduction

The generation of high-fidelity data is a crucial challenge in large astronomical analyses, either for creation of mock sky catalogs in anticipation of future surveys, or in parameter inference pipelines for confrontations with observation. While modeling the complex non-linear physics as gravitational or Magneto-hydrodynamical interactions can be very precise, they cannot be described analytically and generally require expensive numerical simulations. Application of these in traditional statistical inverse problems of astrophysical parameter inference such as the Markov Chain Monte Carlo (MCMC) Metropolis-Hastings algorithm[6] is prohibitively expensive since these Bayesian inference algorithms rely on hundreds of thousands forward model evaluations to estimate posterior probabilities of model parameters.

Recent developments of machine learning techniques enable model parameters inference from experimental data at low computational cost. Progress on emulation techniques based on deep generative models allow high-precision and inexpensive forward models evaluation. Emulation methods in astrophysics, for example [7, 8, 9], are presented as relevant to many problems when a large number of expensive model evaluations is needed and so are those presented in this work.

This project aims at creating a robust emulator based on generative models and neural networks to generate synthetic sky images, composed of galaxy postage stamps. Such emulator will be trained at generating images of galaxies from a number of physical parameters. Meticulous assessments and measurement pipelines will be performed to validate the emulated images beyond visual confirmation.

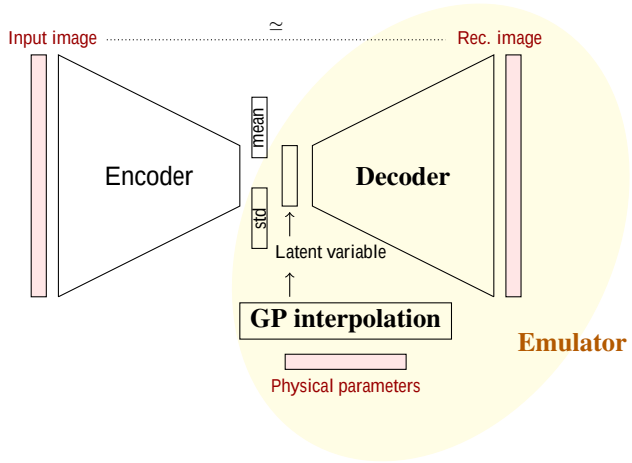

Figure 1: Emulation : an example with Variational Autoencoder and Gaussian Process

Given the complexity involved in real galaxy images, we first create a simulated dataset using GalSim [10], a library for generating images of stars and galaxies. This training set is made of elliptical exponential profile images generated by tuning relatively few parameters on an optimal space-filling sampling scheme [11]. However, deep learning techniques are known to be sensitive to input noise and real galaxy images are much more complex than analytical profiles. Therefore, an emulator trained on this first data wouldn't be able to generate high-fidelity data. Therefore, we define a second dataset made of real galaxy postage stamps from the Cosmic Evolution Survey (COSMOS) [3] taken by Hubble Space Telescope (HST) and related physical parameters given by GalSim and by Leauthaud et al. [4]. Several emulators are then trained and experienced on these datasets. Emulators in this work are designed with two components and trained in two steps, as illustrated Fig.1. First, an unsupervised generative model is trained to reproduce input images through dimensionality reduction and reconstruction (respectively denoted as encoder and decoder Fig. 1), learning a low-dimensional representation of the inputs, known as latent variables. Once trained, the dimensionality reduction network is removed and an interpolator (designated as Gaussian Processes – GP – interpolation Fig. 1) or a conditioning process is trained to map from physical parameters to latent variables. Finally, emulating synthetic galaxy images from physical parameters is presented here as mapping the latter to latent variables and decoding them to images. In this work, we propose three emulator architectures. The first one uses Principal Components Analysis (PCA) as generative model and GP as an interpolator. The second one, as shown in Fig. 1, is composed of a Variational AutoEncoder (VAE) [1] and GP interpolation. The third emulator architecture also consists of a VAE as a generative model but a conditional density estimation network (Masked Autoregressive Flow – MAF [5]) is used as a conditioning process. To assess these emulators performance, we consider several measurement and validation pipelines. We first evaluate reconstruction quality through first order metrics and compare pixel intensities between original and emulated images, compute reconstruction errors as Mean Square Error (MSE) and R-squared error (R^2). Then, we propose higher order metrics to confirm that the emulator is capable of realistic data synthesis. To do so, we measure, on original and emulated images independently, shear parameters using GalSim. The distribution of these parameters should be similar.

This report is organized as follows : in Sect. 2, we describe the 2 datasets used in this work. Emulation methods are detailed in Sect. 3 and results are evaluated in Sect. 4. Finally, perspectives and future work are discussed and conclusions of this work are presented in Sect. 5.

2 Data

2.1 GalSim images

Galaxy images emulated in this work are generated using GalSim [10]. This open-source toolkit aims at providing images of galaxies and stars, including several transformations handling such as dilation, rotation, shear or convolution with Point spread function (PSF) profiles. GalSim allows one to generate and transform a number of parametric profiles or images of galaxies from Cosmic Evolution Survey (COSMOS) [3], a real astrophysical catalog, associated to a PSF model and a parametric representation.

2.1.1 Analytical profiles

Given some well-chosen physical parameters, GalSim is able to generate galaxy profile images. Therefore, as a first simple dataset, we define five physical parameters to tune for galaxy images generation with GalSim. We initially generate a set of parametric exponential profiles. The surface brightness varies along with r as

$$I(r) = \frac{F}{2\ r_0^2} \exp\quad \frac{r}{r_0};$$

with respect to F, total flux and r_0, scale radius. Such generated profiles are circularly symmetric and centered on the coordinate origin. Then we introduce two additional parameters describing a shear distortion to transform a circle into an ellipse with a minor-to-major axis ratio $q = \frac{b}{a}$ and a position angle . In practice, we use as parameters $g_1 = \frac{a\ b}{a+b} \cos 2$ and $g_2 = \frac{a\ b}{a+b} \sin 2$, under the constraint $\sqrt{g_1^2 + g_2^2} < 1$, to define this transformation. The g_1 and g_2 are called "reduced shear" parameters. Finally, these elliptical exponential profiles are convolved with a Gaussian PSF, characterized by its Full Width at Half Maximum (FWHM), linearly related to the variance ^{2} of the related 2-D Gaussian distribution. The values of these 5 physical parameters (F, r_0, g_1, g_2, ^{2}) are generated on a latin hypercube design for optimal space filling. This space-filling scheme is explained in detail in Appendix A.

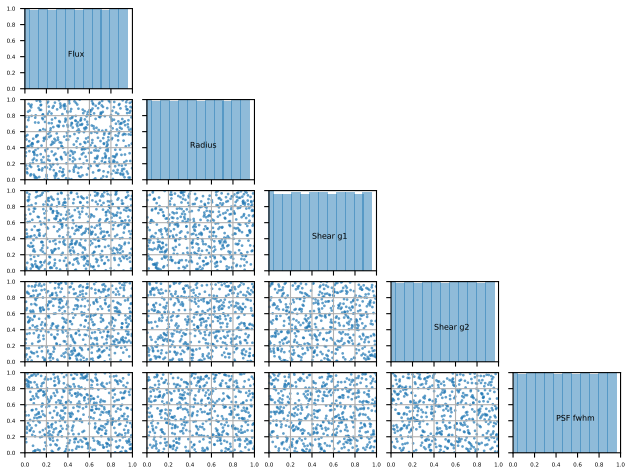

Figure 2: Optimal space filling scheme: latin hypercube for 512 evaluations.

A realization of latin hypercube sampling performed for 512 sampled points is presented Fig. 2. Each graph in the diagonal plots the distribution histogram of each parameter, which is, as expected, uniformly distributed. Each scatter plot represents the normalized value of 2 physical parameters associated to each sampled point. Using these physical parameters samples, we generate with GalSim a representative analytical profiles dataset. Fig. 3 shows a subset of these parametric galaxy images and emphasizes the large diversity of generated elliptical profiles.

2.1.2 COSMOS-type images

The COSMOS survey is the largest survey taken by the Hubble Space Telescope (HST). A publicly available subset of this survey, the COSMOS HST Advanced Camera for Survey (ACS) is used by GalSim as postage stamps provided by [add ref Leauthaud et al. 2007] with a PSF model based on the optical properties of HST. The catalog used here is a catalog of 82000 galaxies taken by the Wide Field Channel (WFC),

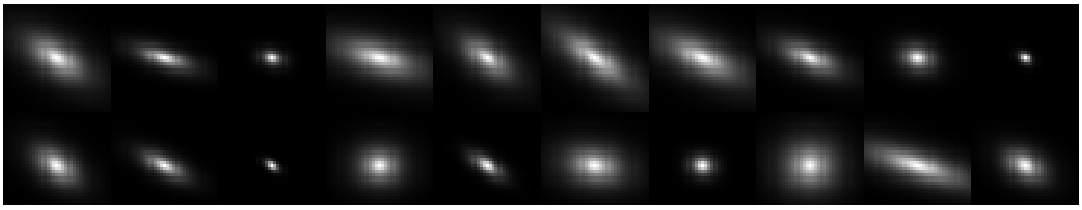

Figure 3: 20 GalSim profiles generated with 5 parameters: galaxy flux, radius, shear profile parameters g_1 and g_2, Gaussian PSF fwhm.

Figure 4: 45 postage stamps (64×64 pixels) of real galaxy images from COSMOS with an absolute depth magnitude lower than $I_{AB} = 25.2$, provided by GalSim.

through the F814W filter with an absolute depth magnitude lower than $I_{AB} = 25.2$. Fig. 4 shows 45 postage stamps of size 64×64 pixels of such real galaxies in COSMOS provided by GalSim. The noisy images of the galaxies depart significantly from the from analytical profiles from Fig. 3, due to the reduction in brightness and features with decreased signal-to-noise ratio. This underlines the fact that analytical profiles are highly over-simplified to understand their complexities of real galaxies.

Every postage stamp in GalSim is associated with a parametric representation. Galaxies have been fitted either with a sersic profile or a bulge and disk profile. The surface brightness of a sersic profile varies as

$$I(r) = \frac{F}{a(n) r_e^2} \exp\left[-b(n)(r/r_e)^{\frac{1}{n}}\right],$$

where r_e is the half-light radius (the radius within half of the galaxy light is contained), $a(n)$ is a normalization function and $b(n)$ computed by GalSim following [add ref Ciotti et al. 1999] with respect to n, the sersic index. The bulge profile is fitted with a DeVaucouleurs profile whose surface brightness is

$$I(r) = \frac{F}{0.010584 r_e^2} \exp\left[-7.66925(r/r_e)^{\frac{1}{4}}\right],$$

and the disk profile is finally fitted with an exponential profile. Similar to elliptical exponential profiles generation (Sect. 2.1.1), the two shear parameters $q = \frac{b}{a}$ and β are added to describe each galaxy. GalSim also provides an estimate of the redshift z of each galaxy. Hence, these 6 parameters (F, r_e, n, q, β, z) represent the physical parameters provided by GalSim. We also include some additional parameters introduced in [add ref Leauthaud et al. 07] : absolute magnitude I_{AB} in the AB system, star formation rate SFR, specific star formation rate sSFR and stellar mass M.

In this set of real galaxy images, the distribution of each parameter is not uniform as in analytical profiles generated in the previous section with an optimal space-filling scheme. Fig. 5 presents the distribution histogram of 5 physical parameters associated to each image (in the diagonal) and scatter plots of normalized values of 2 physical parameters. These plots give an overview of galaxies in our COSMOS-type dataset. The distribution of these 5 physical parameters looks like Gamma or Normal distributions where normalized extreme values (near 0 or 1) are under-represented in the dataset. This might have an influence on the performance of our generative model, which will not be trained enough to generate outliers in the training set, as instance galaxies with a high or medium half-light radius and a high flux.

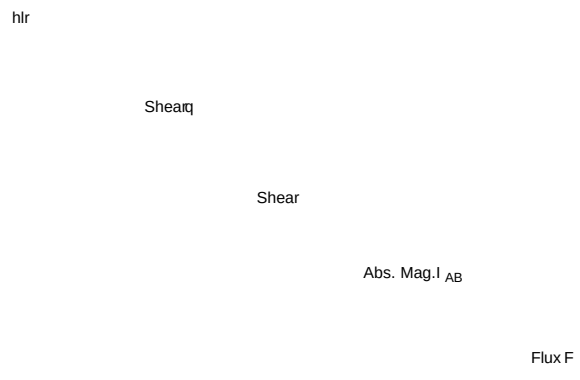

Figure 5: An overview of a few COSMOS galaxy physical parameters given by GalSim et Leauthaud et al. [4]: half-light radius, shear parameters q and β, absolute magnitude and galaxy flux. In the diagonal: the distribution histogram of physical parameters associated to each image. Other plots : scatter plots of normalized values of 2 physical parameters associated to each image.

3 Emulation methods

This section develops different emulation methods build up with a generator (Principal Components Analysis - PCA or Variational AutoEncoder - VAE) and an interpolating (Gaussian Processes - GP) or a conditioning (Masked Autoregressive Flow - MAF) process, their main characteristics and architecture. These emulators are trained and tested on the various datasets we introduced in Sect. 2. Visual results on each dataset and for each method are shown and discussed.

3.1 Emulation with PCA and GP

PCA is known as an orthogonal and linear dimensionality reduction procedure to transform a set of N observations of dimension p expected to be correlated into a smaller set of linearly independent principal components of dimension p. This process performs a M-dimensional ellipsoid fitting to the data. The M principal components are then the axes of the ellipsoid. More formally, a PCA allows to write each observation $X \in \mathbb{R}^p$ as

$$X = Bz, \tag{1}$$

with $B \in \mathbb{R}^{p \times M}$ is the orthogonal truncated basis (i.e. the principal components basis) and $z \in \mathbb{R}^M$ are the weights associated to each principal component. The encoding part of the PCA generator is finding the best ellipsoid fit and thus the principal components matrix B. In practice, the number of principal components M is an hyperparameter and is often set as the smallest number of components that statistically explain at least a certain percentage (99.5 % in Fig. 6 a)) of the variance of the dataset. The decoding part of the PCA generator is basically multiplying this basis matrix B with a latent vector z. As PCA relies on linear assumptions, if the data is not linearly correlated, or if data doesn't fit in an ellipsoid manifold, PCA could fail to find a relevant basis of principal components. Therefore, we expect PCA to be a powerful generator on the simple analytical profiles dataset and quickly be over-performed by other generators for complex COSMOS images.

Once a dimensionality reduction has been performed on the training set and an efficient generator model such as PCA has been trained, we aim at linking the related physical parameters to the low-dimensionality representation of the data, i.e. the latent space represented in Eq. 1 by z. To this aim, we need to observe and ensure a dependency between the observed physical parameters of the training set and the related latent variables. This can be achieved by simply plotting point-wise dependency graphs between 2 dimensions of the latent space and assigning to each point a color corresponding to the value of a specific physical parameter. Such a graph is given Fig. 7 and will be discussed later. If any dependency is noticed, mapping physical parameters $\theta_i \in \mathbb{R}^d$ into a latent variable $z_i \in \mathbb{R}^M$ can be performed with a Gaussian Process (GP) interpolation. GP interpolation process is detailed in Appendix B.

As shown in Fig. 6, we performed a PCA on 512 analytical profiles generated with GalSim in Sect 2.1.1, setting the number M of components to 12 such that the model explains 99.5% of the variance of the data. The resultant 12 components are shown in Fig. 6 b). Fig 7 shows an example of a point-wise dependency plot of the first 2 dimensions of the latent space, i.e. the values of the two first principal components, where a light yellow point corresponds to a large-radius galaxy and a dark blue point to a small-radius galaxy. There is here a very clear dependency between the physical parameter a_0 and the value of the first two components. Similar plots can be derived to ensure that dependencies are observed with every physical parameter and that an interpolation between physical parameters and latent variables is possible. A Gaussian Process is then trained to perform such interpolation and the emulator finally the system {GP, inverse PCA}, with GP interpolating from physical parameters to a latent variable and inverse PCA generating a galaxy image from the interpolated latent variable. Results of this emulator on 10 test-analytical profiles (which are not included in the training set) are shown in Fig. 8. The top row shows the original analytical profiles whereas the bottom row shows the emulated image from the physical parameters associated to each profile in the first row. These results confirm that, visually, our emulator is able to generate realistic simple profiles from physical parameters. However, very sheared galaxies aren't visually well emulated as an artifact appears in the emulated images such that the emulated profile doesn't look like an exponential profile anymore. The last column of Fig. 8 is an example of this phenomenon. Such artefacts occur during image generation by inverse PCA, we train another generative model on this dataset in Sect. 3.2 to fix this issue.

The same PCA pipeline is then applied on the COSMOS-type dataset. The number of components is set to $M = 200$ to explain 97.7% of the variance of the data. The threshold of explained variance ratio is reduced here because a small part of the total variance of the data is related to the noise in training set. We aim here at recovering only the principal components only related to the galaxies in our dataset. Before controlling dependency between physical parameters and latent variable, we need to verify that is able to reproduce more complex images than analytical profiles, Fig. 9. Similarly the first row shows the original COSMOS images. The second row plots the reconstructed image after basis truncation and last row images are noisy versions of second row images, to confirm that reconstructed images with Gaussian noise are visually similar to original COSMOS images. PCA seems to be able to generate realistic galaxies when they are close to simple profiles but fails to reproduce more sophisticated profiles as galaxies flagged with a red star Fig. 9. Moreover, dependencies between physical parameters and latent variable are more difficult to notice in point-wise dependency plots, as illustrated in Fig. 10. Training a Gaussian Process to interpolate physical parameters in such a latent space isn't relevant and emulating COSMOS images with a {GP, inverse PCA} system fails to capture more complex structures. We next explore other methods that might overcome PCA-GP limitation.

3.2 Emulation with VAE and GP

In this section, we explore another generative model, a Variational Autoencoder, associated with Gaussian Processes to emulate more realistic analytical profiles and COSMOS images. VAE and GP emulator is expected to fix or improve PCA issues highlighted in the previous section. A Variational AutoEncoder (VAE) is an unsupervised generative model designed to, given a training data, generate new samples from the same distribution. VAEs are a subset